

Sphero Blueprint Engineering Class Pack

WHAT'S IN THE KIT

- 15 Blueprint Build Kits with beams, plates, gears, axles, and connectors
- Motion components for building working mechanisms
- Storage cases



QUICK-START GUIDE

1. Start with a guided build from Sphero's Blueprint lessons to learn the connector system.
2. Explore gears and gear ratios with a simple crank build.
3. Move to open-ended challenges: students design, build, test, iterate.
4. Sort pieces back into case compartments at cleanup — future you says thanks.

FULL LESSONS IN THIS GUIDE

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PRO TIP: Blueprint's free lesson library on Sphero EDU maps builds to engineering standards.

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LESSON · GRADES 6-8

Gear Ratio Grand Prix

LEARNING OUTCOMES

- Students will build geared mechanisms and explain mechanical advantage
- Students will measure and compare speed vs. torque trade-offs
- Students will iterate a design using recorded ratio data

MATERIALS

- Blueprint Build Kits (1 per team of 2-3)
- Gear ratio recording sheet
- Small weights and string for torque tasks
- Stopwatch or timer

INSTRUCTIONS

1. Warm-up build: a simple hand-crank spinner so teams learn the connector system.
2. Teach gear counting: teeth in ÷ teeth out = the ratio.
3. Challenge A (speed): crank machines race to spin a flag fastest — record ratios used.
4. Challenge B (torque): same machines must lift a weight — record which ratios succeed.
5. Teams chart their data and identify the speed/strength trade-off in their own numbers.
6. Final build: a machine tuned for a surprise task revealed at the end.

ACTIVITY

Gear Ratio Grand Prix — the same machine that wins the speed race fails the lifting task. Students feel mechanical advantage in their hands and see it in their data before anyone defines it.

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LESSON · GRADES 9-12

Mechanism Design Challenge

LEARNING OUTCOMES

- Students will reverse-engineer a mechanism and identify its linkage type
- Students will design and build an original linkage-based device
- Students will document a design process in a professional brief

MATERIALS

- Blueprint Build Kits
- Everyday mechanisms to examine (umbrella, clamp, latch, scissors jack photos or objects)
- Design brief template: problem, sketches, iterations, evaluation
- Calipers or rulers

INSTRUCTIONS

1. Mechanism autopsy: teams sketch how an everyday object converts one motion into another.
2. Introduce linkage vocabulary as they discover it: lever, crank, four-bar, slider.
3. Design sprint: choose a small real problem their own linkage device could solve.
4. Build and iterate — the brief must log every version and why it changed.
5. Peer design review at the midpoint with structured critique.
6. Final: working demo plus the completed design brief, judged on process not polish.

ACTIVITY

Mechanism Design Challenge — from reverse-engineering an umbrella to shipping an original linkage device with a documented design history. The brief is the deliverable; the build is the evidence.